

EMPLOYEE LIFECYCLE AUTOMATION – BACKEND ARCHITECTURE MAP

Note

This is a high-level backend system map.

Employee Lifecycle Engine is a **state-driven, approval-enforced system**, not just workflows.

Final scope, timelines, and cost depend on phase prioritization.

1. CORE SERVICES

A. API Gateway

- Request routing
 - Authentication & Authorization
 - Rate limiting
 - Request logging
 - CORS handling
 - Payload validation
 - Tenant isolation middleware
 - Approval-required action enforcement
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B. Authentication & Authorization Service

- User login / logout
 - JWT token generation
 - Role-based access control (HR, Manager, Employee, Payroll, Legal, IT, Auditor)
 - Session management
 - Secure token refresh
 - Approval authority resolution
 - Audit identity propagation
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2. BUSINESS LOGIC SERVICES

A. Preboarding Orchestration Service

(Triggered only after Hiring Decision = Hire)

Endpoints

POST /api/preboarding/initiate/{candidateId}
GET /api/preboarding/{employeeId}/status
POST /api/preboarding/{employeeId}/complete

Business Logic

- Triggered only from approved hiring decision
 - Convert candidate → employee (soft state)
 - Enforce mandatory preboarding steps
 - Block onboarding until preboarding completion
 - Maintain lifecycle state machine
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B. Offer Letter Service (Approval-Enforced)

Endpoints

POST /api/offers/generate
PUT /api/offers/{id}
POST /api/offers/{id}/submit-approval
POST /api/offers/{id}/approve
POST /api/offers/{id}/reject
POST /api/offers/{id}/send
GET /api/offers/{id}/status

Business Logic

- Generate offer from **approved templates only**
 - Auto-populate role, JD, salary, benefits
 - Editable before submission only
 - Mandatory approval workflow (configurable)
 - Offer immutable after approval
 - Status tracking (Draft / In Approval / Approved / Sent / Accepted / Rejected)
 - Full audit trail
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C. Contract Generation Service

Endpoints

POST /api/contracts/generate
POST /api/contracts/{id}/submit-approval
POST /api/contracts/{id}/approve

POST /api/contracts/{id}/send-signature

GET /api/contracts/{id}/status

Business Logic

- Integrates with Global Labor Law Engine
 - Enforces contract type rules (Permanent / Fixed / Part-Time / Remote)
 - Injects **company mandatory clauses (Policy-as-Data)**
 - Contract versioning & immutability
 - Approval required before e-signature
 - Contract locked after signature
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D. E-Signature & Legal Validation Service

Endpoints

POST /api/contracts/{id}/sign

GET /api/contracts/{id}/signature-status

Business Logic

- Signature allowed only after approval
 - Signature timestamp capture
 - Legal completion gate for employment
 - Signature evidence storage
 - Signature audit logging
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E. Employee Data Collection Service (Privacy-Critical)

Endpoints

POST /api/employees/{id}/personal-data

PUT /api/employees/{id}/personal-data

GET /api/employees/{id}/personal-data

Business Logic

- Secure employee self-service data entry
- Mandatory field enforcement
- Draft & submit workflow
- Field-level encryption
- Role-based data visibility
- Sensitive data masking
- GDPR / SOC2 / ISO27001 compliance logging

F. Onboarding Workflow Service

Endpoints

POST /api/onboarding/initiate/{employeeId}
GET /api/onboarding/{employeeId}/checklist
POST /api/onboarding/{employeeId}/tasks/complete
GET /api/onboarding/{employeeId}/status

Business Logic

- Role-based onboarding checklist generation
- Task assignment to HR / IT / Manager
- Dependency enforcement between tasks
- Orientation scheduling
- Onboarding progress tracking
- Completion gate before probation start

G. Probation & Employment State Service

Endpoints

POST /api/probation/start/{employeeId}
POST /api/probation/{employeeId}/review
POST /api/probation/{employeeId}/confirm
GET /api/probation/{employeeId}/status

Business Logic

- Probation start after onboarding completion
- Probation period rules (country & company specific)
- Manager evaluation capture
- Confirmation / extension / termination decisions
- Employment state transition logging

H. Offboarding Orchestration Service

Endpoints

POST /api/offboarding/initiate/{employeeId}
GET /api/offboarding/{employeeId}/checklist
POST /api/offboarding/{employeeId}/complete

Business Logic

- Exit initiation (voluntary / involuntary)
 - Asset & access revocation checklist
 - Exit interview trigger
 - Dependency on final settlement approval
 - Legal closure enforcement
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I. Final Settlement & Exit Approval Service

Endpoints

POST /api/final-settlement/calculate/{employeeId}

POST /api/final-settlement/{id}/submit-approval

POST /api/final-settlement/{id}/approve

GET /api/final-settlement/{id}/status

Business Logic

- Settlement breakdown (salary, leave, deductions, EOSB)
 - Multi-level approval required
 - Block payroll execution until approval
 - Payslip generated post-approval only
 - Financial audit trail
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J. Unified Approval Workflow Service (Cross-Module)

Endpoints

POST /api/approvals/request

POST /api/approvals/{id}/approve

POST /api/approvals/{id}/reject

GET /api/approvals/{id}/history

Business Logic

- Central approval abstraction
- Configurable approval chains
- Applies to:
 - Offer letters
 - Contracts

- Final settlements
 - SLA tracking & escalation
 - Approval auditability
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3. INTEGRATION SERVICES

A. Recruitment Integration Service

- Candidate → Employee handoff
 - Hiring decision validation
 - Vacancy & JD reference locking
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B. Payroll Integration Service

- Final settlement execution
 - Payroll blocking rules
 - Payslip generation trigger
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C. Global Labor Law Engine

- Country-specific contract rules
 - Probation & notice period enforcement
 - Mandatory clause injection
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D. Document & E-Signature Service

- Offer & contract storage
 - Versioned documents
 - Secure access
 - Legal evidence retention
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E. Notification Service

- Offer sent / accepted
- Contract signature reminders

- Onboarding task alerts
 - Exit & settlement notifications
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4. DATABASE (SUMMARY)

- Employees
 - Employee_Lifecycle_States
 - Offers
 - Offer_Templates
 - Contracts
 - Contract_Versions
 - Contract_Signatures
 - Employee_Personal_Data
 - Onboarding_Checklists
 - Onboarding_Tasks
 - Probation_Records
 - Offboarding_Records
 - Final_Settlements
 - Approval_Requests
 - Approval_Actions
 - Audit_Logs
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DATABASE ARCHITECTURE

A. Employees

- id (PK)
 - candidate_id (FK)
 - jd_id (FK)
 - employment_status (Preboarding / Active / Exiting / Terminated)
 - hire_date
 - termination_date
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B. Employee_Lifecycle_States

- id (PK)
- employee_id (FK)

- state
 - entered_at
 - exited_at
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C. Offers

- id (PK)
 - employee_id (FK)
 - template_version_id
 - salary_data (JSON)
 - benefits (JSON)
 - status
 - approved_at
 - sent_at
 - accepted_at
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D. Contracts

- id (PK)
 - employee_id (FK)
 - country_code
 - contract_type
 - labor_law_version
 - policy_version
 - status
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E. Contract_Signatures

- id (PK)
 - contract_id (FK)
 - signed_at
 - signature_provider
 - evidence_path
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F. Employee_Personal_Data

- id (PK)
 - employee_id (FK)
 - encrypted_payload
 - last_updated
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G. Onboarding_Tasks

- id (PK)
 - employee_id (FK)
 - task_type
 - assigned_to_role
 - status
 - completed_at
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H. Final_Settlements

- id (PK)
 - employee_id (FK)
 - breakdown (JSON)
 - approval_status
 - executed_at
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I. Approval_Requests

- id (PK)
 - entity_type
 - entity_id
 - status
 - requested_by
 - created_at
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J. Audit_Logs

- id (PK)
- user_id
- action

- entity_type
 - entity_id
 - timestamp
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USER STORY COVERAGE

- ☒ EPIC 1 – Preboarding Automation
 - ☒ EPIC 2 – Employee Data Collection
 - ☒ EPIC 3 – Secure Onboarding
 - ☒ EPIC 4 – Probation Tracking
 - ☒ EPIC 5 – Offboarding Automation
 - ☒ EPIC 6 – Final Settlement Approval
 - ☒ Unified Approval Layer (System Rule)
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